

## Multiple Choice

1. **Answer:** C

*Options:* A) 3 units of BMI; B) 0.5 units of BMI; C) 1 unit of BMI; D) 2 units of BMI

*Note:* The correct answer is based on research indicating that vegetarians typically have a lower average body mass index (BMI) compared to meat-eaters, with studies often reporting a difference of approximately 1 unit of BMI.

2. **Answer:** C

*Options:* A) Glucagon increases the rate of glycolysis; B) Glycolysis requires NADP<sup>+</sup>; C) In glycolysis, glucose is cleaved into two three-carbon products; D) The end-product of glycolysis in red blood cells is pyruvate

*Note:* The correct answer is accurate because glycolysis is the metabolic pathway in which one molecule of glucose (a six-carbon sugar) is broken down into two molecules of pyruvate, each containing three carbon atoms, during the process of energy production.

3. **Answer:** D

*Options:* A) Vitamin A; B) Vitamin D; C) Vitamin E; D) Vitamin K

*Note:* Vitamin K is essential for the synthesis of blood clotting proteins, as it plays a crucial role in the carboxylation of certain proteins, enabling them to bind calcium and function properly in the coagulation process.

4. **Answer:** D

*Options:* A) It is never possible to know whether observed effects are caused by the contaminant; B) Human exposure through food is usually much less than exposures that have been studied; C) Controlled animal studies provide more reliable information; D) Reliable dose-response data are difficult to acquire and often unavailable

*Note:* The correct answer highlights that observational epidemiology relies on data from real-world observations rather than controlled experiments, making it challenging to obtain consistent and reliable dose-response data necessary for accurately assessing risk from food contaminants.

5. **Answer:** B

*Options:* A) Increased glucagon secretion in the fasting state.; B) Increased insulin secretion in the fed state.; C) Increased availability of lysine in the fed state.; D) Secretion of cortisol under conditions of stress.

*Note:* Increased insulin secretion in the fed state promotes the uptake of amino acids into cells and stimulates the synthesis of proteins, thereby enhancing the rate of tissue protein synthesis.

## True / False

6. **Answer:** False

*Options:* A) True; B) False

*Note:* The complete oxidation of glucose to carbon dioxide and water typically yields around 30 to 32 molecules of ATP, not 25.

7. **Answer:** False

*Options:* A) True; B) False

*Note:* Low-carbohydrate diets have been shown in various studies to be as effective, if not more effective, for weight loss compared to other dietary approaches, challenging the idea that they were previously deemed less effective.

8. **Answer:** True

*Options:* A) True; B) False

*Note:* The outbreak of mad cow disease raised public awareness and fear about food safety, leading consumers to alter their meat purchasing decisions to mitigate perceived health risks.

9. **Answer:** False

*Options:* A) True; B) False

*Note:* LCD, or low-calorie diet, generally provides more energy than VLCD, or very low-calorie diet, which is designed for rapid weight loss and often restricts caloric intake significantly.

10. **Answer:** True

*Options:* A) True; B) False

*Note:* A sedentary 43-year-old woman weighing 63 kg has a basal metabolic rate and daily caloric needs that align with approximately 1350 kcal per day to maintain her energy balance.

## Long Form Response

11. **Answer:** Obesity is closely linked to an increased risk of endometrial cancer, largely due to the role of hormones, particularly oestrogen. In individuals with obesity, fat cells produce excess oestrogen, which can lead to prolonged exposure of the endometrium, or uterine lining, to this hormone. This excess oestrogen can stimulate the growth of endometrial cells, increasing the likelihood of developing cancer. Therefore, maintaining a healthy weight may help reduce oestrogen levels and lower the risk of endometrial cancer, highlighting the importance of managing obesity for overall health.

*Note:* correct because it clearly explains how obesity contributes to higher levels of oestrogen production, leading to prolonged stimulation of the endometrial lining, which increases the risk of developing endometrial cancer.

12. **Answer:** Type 1 and type 2 diabetes are two distinct forms of diabetes that differ mainly in their causes and management. Type 1 diabetes is an autoimmune condition

where the body's immune system mistakenly attacks and destroys the insulin-producing cells in the pancreas. This leads to a lack of insulin, requiring individuals to rely on insulin therapy for management. In contrast, type 2 diabetes is primarily linked to lifestyle factors and insulin resistance, not autoimmune processes. Treatment for type 2 often includes lifestyle changes and oral medications, while type 1 requires lifelong insulin administration. Understanding these differences is crucial for effective treatment and management of each type.

*Note:* The answer accurately distinguishes type 1 diabetes as an autoimmune disorder characterized by the destruction of insulin-producing pancreatic cells, while identifying type 2 diabetes as primarily related to insulin resistance and lifestyle factors, highlighting their differing pathogenesis and treatment approaches.

*End of Answer Key*